

Losses and Regularization in Machine Learning

Federica Cecchetto

January 21, 2026

What is a loss function?

Setup. Data $(x, y) \sim \mathcal{D}$, model $f_\theta(x)$ (or probabilistic model $p_\theta(y \mid x)$).

Pointwise loss: $\ell(\theta; x, y) \in \mathbb{R}$ measures “how wrong” the model is on one example.

True objective:

$$\mathcal{L}(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}} [\ell(\theta; x, y)].$$

What we optimize:

$$\hat{\mathcal{L}}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(\theta; x_i, y_i).$$

Regularization: optimize $\hat{\mathcal{L}}(\theta) + \lambda \Omega(\theta)$.

Maximum Likelihood Estimation (MLE)

Assume a conditional density $p_{\theta}(y \mid x)$ and i.i.d. data $\{(x_i, y_i)\}_{i=1}^n$.

Likelihood:

$$\mathcal{L}(\theta) = \prod_{i=1}^n p_{\theta}(y_i \mid x_i), \quad \textbf{Log-likelihood:} \quad \ell(\theta) = \sum_{i=1}^n \log p_{\theta}(y_i \mid x_i).$$

MLE:

$$\hat{\theta}_{\text{MLE}} \in \arg \max_{\theta} \sum_{i=1}^n \log p_{\theta}(y_i \mid x_i).$$

Intuition: pick parameters that make the observed data as probable as possible under the model.

Cross-Entropy (CE) and Kullback-Leibler (KL) divergence

Let P be the true distribution of Y (or $Y \mid X = x$) and Q a model distribution.

$$\text{CE}(P, Q) = \mathbb{E}_{y \sim P}[-\log Q(y)] = -\sum_y P(y) \log Q(y).$$

Assume P is absolutely continuous w.r.t. Q ($P \ll Q$). Then

$$\text{KL}(P \parallel Q) = \mathbb{E}_{y \sim P} \left[\log \frac{P(y)}{Q(y)} \right] = \text{KL}(P \parallel Q) = \sum_y P(y) \log \frac{P(y)}{Q(y)}.$$

Cross-Entropy vs KL Divergence

Distributions:

$$p = (0.5, 0.5, 0), \quad q = (0.9, 0.1, 0)$$

Cross-Entropy

$$H(p, q) = - \sum_i p_i \log q_i = -[0.5 \log 0.9 + 0.5 \log 0.1] \approx 1.204$$

$$H(q, p) = - \sum_i q_i \log p_i = -[0.9 \log 0.5 + 0.1 \log 0.5] = -\log 0.5 \approx 0.693$$

KL divergence

$$\text{KL}(p\|q) = \sum_i p_i \log \frac{p_i}{q_i} = 0.5 \log \frac{0.5}{0.9} + 0.5 \log \frac{0.5}{0.1} \approx 0.511$$

$$\text{KL}(q\|p) = \sum_i q_i \log \frac{q_i}{p_i} = 0.9 \log \frac{0.9}{0.5} + 0.1 \log \frac{0.1}{0.5} \approx 0.368$$

CE and KL are the same objective up to a constant

Start from:

$$\text{CE}(P, Q) = \mathbb{E}_P[-\log Q(Y)].$$

Add and subtract $-\log P(Y)$:

$$\text{CE}(P, Q) = \mathbb{E}_P[-\log P(Y)] + \mathbb{E}_P\left[\log \frac{P(Y)}{Q(Y)}\right].$$

Therefore:

$$\text{CE}(P, Q) = H(P) + \text{KL}(P\|Q),$$

where $H(P) = \mathbb{E}_P[-\log P(Y)]$ is the (Shannon) entropy of P .

Consequence: since $H(P)$ does not depend on Q ,

$$\arg \min_Q \text{CE}(P, Q) = \arg \min_Q \text{KL}(P\|Q).$$

Why Maximum Likelihood equals Cross-Entropy minimization

Setup.

$$(x, y) \sim \mathcal{D}, \quad P(y \mid x) \text{ true conditional}, \quad Q_{\theta}(y \mid x) \text{ model}$$

Maximum Likelihood (population objective):

$$\theta^* \in \arg \max_{\theta} \mathbb{E}_{(x, y) \sim \mathcal{D}} [\log Q_{\theta}(y \mid x)]$$

Equivalent minimization:

$$= \arg \min_{\theta} \mathbb{E}_{(x, y) \sim \mathcal{D}} [-\log Q_{\theta}(y \mid x)]$$

Law of total expectation:

$$= \arg \min_{\theta} \mathbb{E}_{x \sim \mathcal{D}_X} \left[\mathbb{E}_{y \sim P(\cdot \mid x)} [-\log Q_{\theta}(y \mid x)] \right]$$

Definition of cross-entropy:

$$= \arg \min_{\theta} \mathbb{E}_{x \sim \mathcal{D}_X} [\text{CE}(P(\cdot \mid x), Q_{\theta}(\cdot \mid x))]$$

Regression losses: MBE, MAE, MSE

Given targets $y_i \in \mathbb{R}$ and predictions $\hat{y}_i = f_{\theta}(x_i)$, residuals $r_i = \hat{y}_i - y_i$.

Mean Bias Error (MBE):

$$\text{MBE} = \frac{1}{n} \sum_{i=1}^n r_i$$

Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |r_i|$$

Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n r_i^2$$

MSE is Gaussian maximum likelihood

Assume a regression likelihood:

$$Y \mid X = x \sim \mathcal{N}(\mu = f_{\theta}(x), \sigma^2) \quad \text{with fixed } \sigma^2.$$

Then for one sample:

$$-\log p_{\theta}(y \mid x) = \frac{1}{2\sigma^2}(y - f_{\theta}(x))^2 + \frac{1}{2}\log(2\pi\sigma^2).$$

Summing over i :

$$\sum_{i=1}^n -\log p_{\theta}(y_i \mid x_i) = \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - f_{\theta}(x_i))^2 + \text{constant}.$$

Therefore: maximizing Gaussian likelihood $\iff$ minimizing MSE.

Regularization: definition and motivation

Empirical risk minimization:

$$\min_{\theta} \hat{\mathcal{L}}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(\theta; x_i, y_i)$$

Regularized objective:

$$\min_{\theta} \hat{\mathcal{L}}(\theta) + \lambda \Omega(\theta), \quad \lambda > 0$$

Why regularization?

- Finite data $\Rightarrow$ many models fit the training set equally well
- Some solutions are overly complex or unstable

Effect of regularization:

- penalizes complex or undesirable solutions
- reduces overfitting and improves generalization
- stabilizes optimization and selects among equivalent minima

Ridge and Lasso regularization

Ridge (ℓ_2):

$$\Omega(\theta) = \|\theta\|_2^2 \quad \Rightarrow \quad \min_{\theta} \hat{\mathcal{L}}(\theta) + \lambda \|\theta\|_2^2 \quad \Longleftrightarrow \quad \min_{\theta} \hat{\mathcal{L}}(\theta) \text{ s.t. } \|\theta\|_2^2 \leq t$$

Lasso (ℓ_1):

$$\Omega(\theta) = \|\theta\|_1 = \sum_j |\theta_j| \quad \Rightarrow \quad \min_{\theta} \hat{\mathcal{L}}(\theta) + \lambda \|\theta\|_1 \quad \Longleftrightarrow \quad \min_{\theta} \hat{\mathcal{L}}(\theta) \text{ s.t. } \|\theta\|_1 \leq t$$

- Ridge: shrinks weights smoothly toward zero
- Lasso: promotes sparsity (exact zeros), feature selection

Regularization as Bayesian MAP estimation

Bayes rule:

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta) p(\theta)}{p(\mathcal{D})}$$

Maximum a posteriori (MAP):

$$\theta_{\text{MAP}} = \arg \max_{\theta} \log p(\mathcal{D} \mid \theta) + \log p(\theta)$$

Equivalently:

$$\theta_{\text{MAP}} = \arg \min_{\theta} \left[-\log p(\mathcal{D} \mid \theta) - \log p(\theta) \right]$$

Interpretation: regularization term is a negative log-prior.

Gaussian prior leads to Ridge regularization

Assume a Gaussian prior:

$$p(\theta) = \mathcal{N}(0, \sigma^2 I)$$

Then:

$$-\log p(\theta) = \frac{1}{2\sigma^2} \|\theta\|_2^2 + \text{const}$$

MAP objective:

$$\min_{\theta} \left[-\log p(\mathcal{D} \mid \theta) \right] + \frac{1}{2\sigma^2} \|\theta\|_2^2$$

Conclusion: Gaussian prior $\iff$ Ridge regularization.

Laplace prior leads to Lasso regularization

Assume a Laplace prior (i.i.d. components):

$$p(\theta_j) = \frac{1}{2b} \exp\left(-\frac{|\theta_j|}{b}\right)$$

Joint prior:

$$p(\theta) \propto \exp\left(-\frac{1}{b} \|\theta\|_1\right)$$

Negative log-prior:

$$-\log p(\theta) = \frac{1}{b} \|\theta\|_1 + \text{const}$$

Conclusion: Laplace prior $\iff$ Lasso regularization.

Logits, softmax, and cross-entropy

Logits: $z = (z_1, \dots, z_K) \in \mathbb{R}^K$ and **Softmax probabilities:**

$$q_i = \text{softmax}(z)_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}, \quad \sum_{i=1}^K q_i = 1.$$

Target distribution: $y = (y_1, \dots, y_K)$ with $y_i \geq 0$, and $\sum_{i=1}^K y_i = 1$.

Cross-entropy loss:

$$\text{CE}(y, q) = - \sum_{i=1}^K y_i \log q_i.$$

Using softmax:

$$\text{CE}(y, \text{softmax}(z)) = - \sum_{i=1}^K y_i z_i + \log \left(\sum_{j=1}^K e^{z_j} \right).$$

Why softmax + cross-entropy gives simple gradients

Define the loss as

$$\mathcal{L}(z) = \text{CE}(y, q) = - \sum_{i=1}^K y_i \log \text{softmax}(z)_i = - \sum_{i=1}^K y_i z_i + \log \left(\sum_{j=1}^K e^{z_j} \right).$$

Differentiate w.r.t. each logit z_i :

$$\frac{\partial \mathcal{L}}{\partial z_i} = -y_i + \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}} = q_i - y_i.$$

$$\boxed{\frac{\partial \text{CE}(y, \text{softmax}(z))}{\partial z_i} = q_i - y_i}$$